# PREDICTING THE SHAPE OF THE THAR DESERT USING DEEP LEARNING

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*Abstract*—**Reliable forecasts of desert expansion are needed at the start of the planning cycle to prioritize land restoration, protect rain-fed agriculture, and guide water and infrastructure investments in the Thar Desert region. This paper proposes deep learning techniques applied to satellite imagery to predict the future expansion of the Thar Desert under seasonal and interannual variability. Using Google Earth Engine datasets, particularly MODIS and Dynamic World, we benchmark a traditional Cellular Automata (CA)-Markov approach against Neural Cellular Automata (NCA), PredRNN, and a Transformer–Convolutional Cellular Automata (TC-CA) hybrid model to capture spatio-temporal dynamics. CA-Markov achieves a Lee–Sallee Index of 0.8643 and Figure of Merit of 0.2516; NCA attains 0.861 overall accuracy, 0.686 F1, and 0.707 Kappa; PredRNN records an SSIM of 0.895633 and pixel-wise correlation 0.9595; and TC-CA delivers the strongest classification performance with 90.62% accuracy and a Kappa of 0.7983. These results provide an actionable, data-driven basis for policy interventions aimed at mitigating desertification and increasing climate resilience in the region.**

## I. INTRODUCTION

Desertification, a major environmental issue in arid and semi arid regions in particular [1], affects roughly 500 million people globally, by contributing to poverty, food insecurity, and water scarcity [2]. The Thar Desert, which spans across parts of India and Pakistan, is particularly vulnerable because of its relatively high population density of 84 persons per square kilometer and a fragile ecosystem [3]. Despite getting only $100 - 500$ mm of rainfall annually, the region is home to one of the highest desert population densities, with 70% depending on rain-fed agriculture and pastoralism. However, depletion of groundwater and drastic seasonal shifts influenced by climate change greatly threaten its arability, increasing the threat of desertification [4].

While remote sensing has enabled effective monitoring of desertification, the task of prediction remains a challenge. Traditional Machine Learning methods, such as Random Forests and decision trees, rely on manually engineered features, restricting their ablity to learn complex patterns. For example, Li et al. [5] used Random Forests to achieve 87.2% accuracy for desertification mapping, but their reliance on handcrafted features limited the generalization of their results. Likewise, Cellular Automata (CA) have dominated the study for LULC simulation [6] [7]. Traditional CA-Markov models calculate transition probabilities over a time period to predict changes between land-use classes. While they may be effective, these models are spatially limited, as transition probabilities do not account for neighborhood effects [8]. Deep Learning models, such as Convolutional Neural Networks, attempt to address this limitation as they excel at spatial feature extraction. U-Net, for example, has become a standard for semantic segmentation in remote sensing, creating accurate boundaries, which is essential for desertification modeling [9]. However, they fail to account for the temporal dependencies.

To address this gap, hybrid and generative models have been used in recent research. Architectures like ConvLSTM and Graph Neural Networks (GNNs) have somewhat successfully predicted LUCC by factoring in both spatial and temporal information[10]. Hybrid models, such as the TC-CA (Transformer-Convolutional Cellular Automata) combine the strengths of multiple architectures to predict LULC changes. Specifically, a CNN is used to extract the spatial features, Transformer to model temporal dependencies, and then finally a CA module for spatially informed predictions[11]. Additionally, Neural Cellular Automata (NCA) learn the update rules dynamically from data rather than depending on any predefined and static spatial rules, and has achieved remarkable results which capturing the dynamics of desertification [12]. Integrating CNNs with CA, (also known as CA-ANNs) (Artifical Neural Networks), allows learning neighborhood effects from data, improved spatial prediction accuracy through ingrained landscape fragmentation patterns, especially for sparse desert areas.

In this study, we present a comparative analysis of these deep learning paradigms to forecast desertification in the Thar Desert. We integrate multi-decadal MODIS temporal data with high-resolution Dynamic World land cover probabilities to train and evaluate four distinct architectures: a baseline CA-Markov, a generative Neural Cellular Automata (NCA), a Spatiotemporal ConvLSTM (PredRNN), and a hybrid TC-CA model. To the best of our knowledge, this work represents the first application of these advanced generative and hybrid models specifically to the Thar region. By evaluating their ability to capture both seasonal variability and long-term expansion, we aim to provide robust forecast maps that can guide policymakers in designing effective climate interventions and sustainable land management practices.

## II. Methodology

### A. Area of Interest

To define the scope of the research, an area of interest (AOI) is defined marking the region under focus for studying the trends of desertification. This is done by taking an existing map of Thar Desert from Terrestrial Ecoregions defined by the World Wildlife Fund (WWF) [13] for 2012. A bounded box, spanning is made over the region by taking the maximum limits of the desert in all directions. It is added to leave a buffer for any desertification (desert expansion) activity at the boundaries of the desert where it is most likely to occur. Fig. 1 below shows the AOI defined in Google Earth Engine which was used to generate the map file.

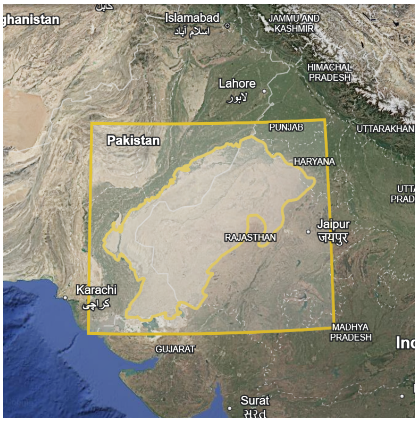


Fig. 1. Area of Interest over Thar Desert

The AOI is partitioned into a regularized grid of tiles measuring $0.1^o$ x $0.1^o$ degrees (approximately 10km x 10km at the equator). Each tile is assigned a unique (i,j) coordinate, representing its position in a 2D spatial matrix. This grid acts as a 'canvas' upon which all subsequent data layers are projected, ensuring perfect spatial alignment between spectral and categorical data.

### B. Datasets

Two primary datasets were accessed via the Google Earth Engine API:

*1) MODIS:* The *MOD09A1.061 Terra Surface Reflectance 8-Day Global 500m* is a global data product under the MODIS (Moderate Resolution Imaging Spectroradiometer) collection coming from the Terra satellite which was launched in 2000. It provides raw surface reflectance band values obtained at a 500m pixel resolution. These bands correspond to different wavelengths of the electromagnetic spectrum [14] Table I shows these raw bands.

TABLE I
MODIS Data

| Band | Wavelength | Colour |
|---|---|---|
| `sur_refl_b01` | 620-670nm | RED |
| `sur_refl_b02` | 841-876nm | NIR |
| `sur_refl_b03` | 459-479nm | BLUE |
| `sur_refl_b04` | 545-565nm | GREEN |
| `sur_refl_b05` | 1230-1250nm | TIRS1 |
| `sur_refl_b06` | 1628-1652nm | SWIR1 |
| `sur_refl_b07` | 2105-2155nm | SWIR2 |

Instead of passing raw reflectance bands, an additional preprocessing step is involved: the raw MODIS band values are scaled to their original values and transformed into four distinct spectral indices. These indices act as 'hand-crafted features'; enhancing the receptibility of the models to physical sensitivies like surface moisture, vegetation vigor, and soil composition, reducing the high-dimensional spectral noise into physically meaningful values.

The following formulae were applied to the bands for obtaining the normalized, mean indices corresponding to each subtile [15].

- **NDVI (Normalized Difference Vegetation Index):** quantifies photosynthetic activity and biomass.

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

- **MSAVI (Modified Soil Adjusted Vegetation Index):**quantizes vegitative activity, but minimizes the impact of soil brightness in areas with sparse vegetation.

$$MSAVI = \frac{2NIR + 1}{2} - \frac{\sqrt{(2NIR + 1)^2 - 8(NIR - \text{Red})}}{2}$$

- **TGSI (Tasseled Cap Greenness Index):** captures sensitivity to topsoil texture and desertification.

$$TGSI = \frac{Red - Blue}{Red + Blue + Green}$$

- **Albedo:** measures the total solar radiation reflected by the surface.

$$Albedo = 0.356 \times Blue + 0.13 \times Red + 0.373 \times NIR$$

$$+ 0.085 \times SWIR1 + 0.072 \times SWIR2 - 0.0018$$

*2) Dynamic World:* Dynamic World V1 is a 10m near-real-time (NRT) dataset derived from Sentinel-2. It provides Land Use/Land Cover (LULC) labels and probabilities for nine classes [16].

Table II shows 9 classes obtained from Dynamic World:

These probabilities allow for models to detect and learn subtle land-cover shifts rather than 'hard' classifications. However, due to the unavailability of ground surveys, here they are also used as rough ground truth labels to observe and distinguish the tiles between desert and non-desert, with the assumption that the shrub & scrub and bare classes are the most relevant for desert presence.

TABLE II
DYNAMIC WORLD DATA

| Classes |
| --- |
| water |
| trees |
| grass |
| flooded_vegetation |
| crops |
| shrub_and_scrub |
| built |
| bare |
| snow_and_ice |

### C. Data Processing

The extracted data is transformed into a model-ready tensor after the following steps ensuring spatio-temporal synchronization. This entire process is summarized in Fig. 2:

- **Temporal Synchronization:** Owing to the differing revisit cycles of MODIS and Dynamic World, the MODIS 8-day schedule is adopted as the master temporal reference (T). The revisit frequency of Sentinel-2 is between 2-5 days depending on latitude, therefore for every MODIS timestamp, a corresponding Dynamic World composite is generated by averaging all available DW scenes within a centered 8-day window ([t-3,t+4] days).
- **Spatial Aggregation:** A zonal mean reducer is used to aggregate high-resolution pixels into $0.1^o$ titles. For MODIS-derived indices, this corresponds to averaging 400 pixels per tile. On the other hand, for Dynamic World, it serves as a way for rudimentary downsampling, by averaging 1,000,000 pixels per tile to conserve the same spatial resolution while preserving the statistical mean of the area.
- **Interpolation:** Due to cloud cover on several days, both Sentinel 2 and MODIS satellite imagery contained missing values, a common issue in optical remote sensing. To address this, a two stage linear interpolation strategy was adopted, consistent with approaches reported by De Luca et al. [17] and Tran et al. [18].
  First, temporal interpolation was performed independently for each spatial tile by linearly interpolating missing values across consecutive dates within the same tile. Subsequently, any remaining missing values were filled using spatial interpolation by computing the mean across all valid tiles for the same date, ensuring spatial consistency while avoiding the introduction of artificial temporal patterns.
- **Normalization:** Following interpolation, the MODIS vegetation indices were normalized using min–max normalization to map all values into the range $[0, 1]$. For each vegetation index, the minimum and maximum values were computed across the entire spatio-temporal extent of the dataset.
- **Tensor Formation:** After preprocessing, 4D spatio-temporal tensors were constructed for MODIS and Dynamic World data replicating Chouikhi [15]. In both cases, $T$ denotes the temporal dimension (timestamps), while $W \times H$ represent the spatial dimensions of the AOI. For MODIS the data was organized into a of shape $(T \times W \times H \times 4)$, corresponding to the four normalized vegetation indices. Similarly, the Dynamic World data were arranged into a tensor of shape $(T \times W \times H \times 9)$ corresponding 9 classes. The tensors were stacked chronologically while maintaining spatial consistency, enabling independent yet aligned spatio-temporal modelling of vegetation dynamics and land cover probabilities.

The completed tensor is fed to four different models, each designed to capture a different aspect of the landscape's evolution, with some modifications.

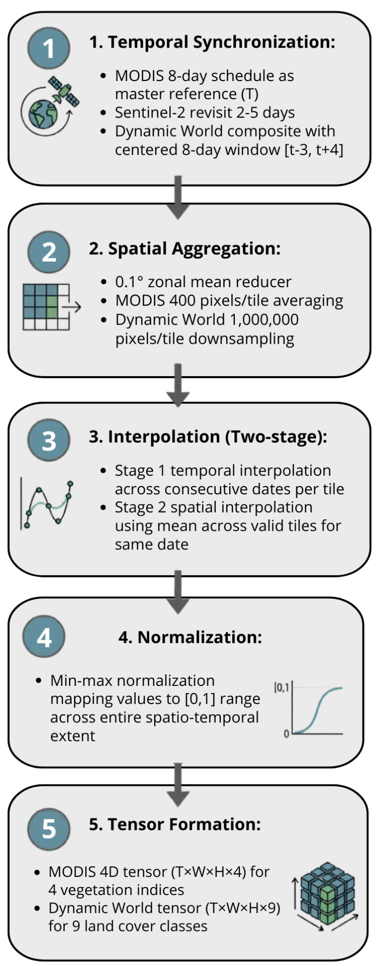


Fig. 2. Overview of the data pipeline used to transform satellite observations into model-ready spatio-temporal tensors.

### D. Models

For this research, four different models have been implemented: CA-Markov, Neural Cellular Automata (NCA), PredRNN, and the TC-CA hybrid model. These architectures model both spatial and temporal dynamics of land-cover change under different inherent assumptions. The focus is to predict desertification patterns in the Thar Desert using LULC data from DynamicWorld.

The following subsections outline the methodology for each model.

*1) CA Markov:* Cellular Automata Markov is a technique widely used for spatiotemporal analysis. It was used to predict the evolution of rocky desertification in China [8]. While Markov's transition matrix deals with temporal changes over the year, Cellular Automata considers the neighbouring tiles and their influence. While Qian [8] used the data after every 5 years, here Dynamic World data is used yearly from 2016 till 2024. A particular set of pre-moonsoon days, were selected in order to retain the sesonality and not confuse the model as suggested by [19]. The data is then averaged out for each year, and the 4D tensor is reduced to 3D tensor by assigning the label of highest class per tile instead of using probabilities.

The initial stage included quantifying the temporal change by calculating a Markov Transition Matrix for each pair of consecutive years upto the 8th year, and the last year is kept for evaluation. The model computed a $K \times K$ transition probability matrix where $K = 9$ represents 9 classes of Dynamic World, and each entry $T[i, j]$ represents the probabilty of class i turning into class j in the next year.

To include the spatial behaviour, Cellular Automata was applied by performing a convolution with a kernel of size $3 \times 3$ to count the number of neighbouring pixels belonging to the same class. This was converted into class wise local probabilities, forming a neighbourhood suitability matrix. This served as a replacement for the original suitability matrix used by Qian [8] which used parameters such as temperature and other indices.

An acceleration matrix was applied to fine-tune the base transition matrix which helps to modify the transition intensities between classes, but the data is mostly stable so the matrix has no effect on the model.

After the improved transition matrix and suitabilty matrix have been calculated, the final score is calculated using the following equations where $\lambda$ was taken as 0.6.

$$\text{score}_j(p) = \lambda\, N_j(p) + (1-\lambda)\, T_{ij}^{\text{improved}}$$

$$\text{new\_class}(p) = \arg\max_j\ \text{score}_j(p)$$

The tiles are labelled by the classes with the maximum probability corresponding to each of them.

At the end of the implementation, the results are evaluated based on the prediction of the previous year, whereas the simulation proceeds year by year. This iterative process makes sure that rather than jumping to the final state, the evolution of spatial patterns remains realistic.

*2) Neural Cellular Automata (NCA):* Neural Cellular Automata are generative models that leverage deep learing to learn the update rules of cellular automata. This learning allows the model to dynamically adjust adapt at each timestamp, allowing for modelling of complex emergent behaviours. This model has found applications in tissue morphogenesis and image synthesis, and is now being used for LULC simulation [12].

The model was implemented in a approach similar to that taken by Jinian Zhang, and Lanfa Liu, for land use changes simulation in Wuhan, China. The initial cell state tensor $x$ is of shape $(B, C, H, W)$, where $B$ is batch size, $C$ is the number of state channels (features per cell), and $H, W$ are the grid dimensions, which is passed to the model architecture as follows:

- **Perception Module:** extracts local neighbourhood information of gradients by applying a series of $3 \times 3$ Sobel Filters
- **Rule Network:** a Convolutional Neural Network feeding on the local context and state to determine how much each cell should change. GroupNorm is applied over the input, followed by ReLU activation. Intermediate layers with pointwise convolution are used to further process the features and mix information per cell across the channel dimension, enhancing the network's capacity. Each of these is followed by Layer Normalization and ReLU. The Output Conv $1 \times 1$ maps the hidden features back to the original number of state channels ($state\_ch$).
- **Update Loop:** The output tensor $\delta$ (delta) of shape $(B, C, H, W)$ from the rule network represents the suggested change (delta) to be applied to the current cell state. It applies the update rule after clipping and stochastic masking.

Here the model is fed a 4D tensor input of $(T, W, H, 13)$ with 13 channels representing the 9 Dynamic World classes, and 4 vegetation indices of NDVI, MSAVI, TGSI and Albedo from MODIS.

*3) PredRNN - Spatiotemporal ConvLSTM:* The ConvLSTM model is widely used to learn and model changes over space and time. Here, the model is used is the PredRNN or the SpatioTemporal ConvLSTM, which is an enhancement over the traditional ConvLSTM model and capable of capturing more complex patterns by relying on dual memory states to capture complex motion and long-range dependencies in sequential imagery, as opposed to single temporal memory in ConvLSTM.Several of these SpatioTemporal ConvLSTM layers are stacked one after another at each time step. It alternates between teacher-forcing for observed frames and autoregressive generation for predicted frames. For desertification prediction, the model is trained on the Dynamic World input 4D data tensor and predicts after every 12 frames.

*4) TC-CA Hybrid:* The TC-CA (Transformer-Convolutional Cellular Automata) model, as introduced by Li et al. [11], combines Convolutional Neural Networks (CNN), Transformer architecture, and Cellular Automata (CA) to simulate land-use land-cover change (LUCC). By integrating spatial and temporal features, it overcomes the limitations of traditional models that neglect temporal dependencies, allowing for more accurate LUCC predictions.

- **CNN:** extracts spatial features from land-use maps to produce a compact representation.
- **Transformer:** captures temporal features in the LUCC process via self-attention.

- **Cellular Automata (CA) Transition Rule:** determines the future state of each cell depending on its spatiotemporal context.

## III. Results

### A. CA Markov

The results have been evaluated based on the prediction of 9th year for which the ground truth is available. They include the Lee-Sallee Index which is the ratio of correctly predicted cells divided by total number of cells

$$\mathrm{LS} = \frac{|A_0 \cap A_1|}{|A_0 \cup A_1|},$$

Where $A_0$ = observed map and $A_1$ = predicted map. Our model achieved a score of 0.8643.

A second metric of evaluation used is Figure of Merit (FOM) which evaluates the model's ability to correctly predict the changed pixel. An FOM of only 0.2516 is acheived, suggested high dominanace of false positives in the predictions.

### B. Neural Cellular Automata

The Neural Cellular Automata model has been trained using Adam Optimizer and focal loss to give importance to less common classes. It shows promising results for predicting the next state of LULC for the desert. These results were evaluated on test data using three metrics: Overall Accuracy, F1 Score and Kappa Coefficient.

Setting the iteration time to 1 yielded the best results. Increased iteration time degraded the results and have been excluded [12].

TABLE III
NEURAL CELLULAR AUTOMATA EVALUATION METRICS

| Metric | Value |
|---|---|
| Overall Accuracy | 0.861 |
| F1 Score | 0.686 |
| Kappa | 0.707 |

### C. PredRNN - Spatiotemporal ConvLSTM

The perfomance of the PredRNN model is assessed using complementary evaluation metrics such as:

- Mean Squared Error (MSE): 0.001011
- Structural Similarity Index (SSIM). SSIM evaluates perceptual similarity by comparing structural information, luminance, and contrast between the prediction and ground truth. Values range from $-1$ to 1, with 1 indicating perfect structural similarity. The model achieved a high SSIM score of **0.895633**
- Pixel-wise Correlation.The model achieved a high overall pixel-wise correlation of **0.9595**.

These metrics demonstrate that the model achieves low reconstruction error, high structural similarity, and strong pixel-level correlation with the ground truth.

### D. TC-CA Hybrid

The TC-CA model shows robust performance in identifying spatio-temporal changes within the LULC dataset. The training history, including accuracy and the custom Figure of Merit (FoM), is shown in Figure 3.

It achieved a final accuracy of 90.62% on the test set, indicating a strong agreement between predicted and actual changes. Additionally, a Kappa score of 0.7983 indicates that the model's predictions align with the ground truth far beyond random chance.

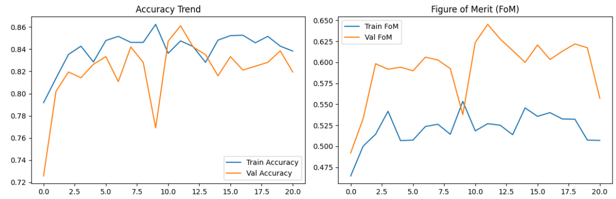


Fig. 3. Accuracy Trend and Figure of Merit (FoM) during model training. The plots illustrate the model's learning progression on both training and validation datasets.

In addition to the quantitative metrics, the TC-CA model generated a future suitability and transition prediction map. Figure 3 illustrates the model's prediction for the next time step.

## IV. Discussion

In this study, four complementary approaches have been developed and compared for forecasting the desertification dynamics in Thar Desert using multi-decadal MODIS indices and Dynamic World land cover probabilities. By combining traditional CA-Markov modelling with Neural Cellular Automata, PredRNN, and the TC-CA hybrid architecture, it is demonstrated that learning models can be used to capture both the seasonal variability and longer term expansion patterns of desert land cover. The CA-Markov baseline provides a spatially coherent reference simulation, while the Neural CA demonstrates the capacity of learned update rules to reproduce complex local transitions. The PredRNN model demonstrates minimal reconstruction error and high structural similarity, showing strong alignment with the observed probability fields. The TC-CA hybrid achieving an accuracy of 90.62% and a Kappa score of 0.7983, highlights the benefits of jointly modelling spatial context and temporal dependencies. These results suggest that the integration of spatiotemporal deep learning with remote-sensing products can be used to develop early-warning systems as well as evidence-based land management and policy decisions in such highly vulnerable dryland regions.

Building from this study, the models can be made more effective by training on higher-resolution data such as that from Landsat (30 m). Other than satellite imagery, including socioeconomic factors and drivers like GDP, population growth rate and deforestation rates may allow better land degradation modelling. Expanding the definition of "desert" beyond only the bare and shrub-scrub classes can further fine-tune the results.

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